SOVEREIGN AUTHORITY EDITION · 2026 · CANADIAN PATENT NO. CA 3,300,102

Global Record Governance Framework

Complete Institutional Project Proposal

Inventor-Controlled Stewardship Model

INVENTOR & FRAMEWORK ARCHITECT

Tarek Wahid

Canadian Patent No. CA 3,300,102

CLASSIFICATION: INSTITUTIONAL · CRP v1.0 · HASH-SEALED · TRILINGUAL (EN/FR/AR)

FRAMEWORK	PATENT	VALUE	STANDARDS
Six-Layer Deterministic Governance Architecture	CA 3,300,102 Invented & Owned	$27.1B – $206.8B 10-Year Projection	ISO · OECD · UN 12+ Frameworks

Executive Summary

The Global Record Governance Framework (GRGF) introduces a paradigm shift in institutional governance infrastructure. It is the world's first deterministic, append-only governance record system designed to make institutional actions provable, omissions visible, and authority chains verifiable. This is not a registry, workflow engine, or analytics platform — it occupies a unique position as governance record-of-record infrastructure.

THE CORE INNOVATION

- Omission-Awareness: The first framework to treat the absence of action as a recordable, auditable governance event.
- Deterministic Denial Logic: Every decision — approval or denial — is cryptographically sealed with full chain-of-custody.
- Anti-Capture Clauses (AC-01 to AC-05): Legal safeguards preventing corporate or political override of framework neutrality.
- Six-Layer Architecture: Ingestion, Validation, Cryptographic Proof, Policy Engine (OPA/Rego), Audit, and Federation layers.
- Inventor-Controlled Stewardship: Framework governance is neutral and inventor-controlled, not vendor-dependent.

PROJECTED IMPACT

Scenario	5-Year	10-Year	SROI
Conservative	$4.2B	$27.1B	8:1
Moderate	$12.8B	$118.6B	14:1
Aggressive	$28.4B	$206.8B	20:1

Based on 0.3% integrity improvement threshold applied to $2.6T annual governance loss globally. Conservative scenario assumes Canada-only deployment; aggressive assumes G20 federation by 2035.

Six-Layer Deterministic Architecture

LAYER 1 INGESTION

Structured intake of governance events via validated schemas (OpenAPI 3.1). Supports bulk, API, and manual entry with mandatory metadata.

LAYER 2 VALIDATION

Schema enforcement, authority verification, and completeness checks. Rejects malformed or unauthorized submissions deterministically.

LAYER 3 CRYPTOGRAPHIC PROOF

Merkle transparency logs with three proof types: Inclusion, Consistency, and Audit. HSM-backed key management ensures non-repudiation.

LAYER 4 POLICY ENGINE

Open Policy Agent (OPA) with Rego policy language. Deterministic policy evaluation — same input always produces same output.

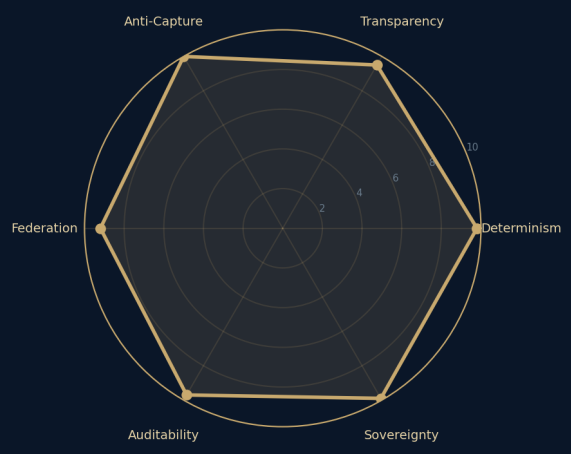
LAYER 5 AUDIT & COMPLIANCE

Continuous audit trail with positive and negative verification. Records of omission are first-class audit events.

LAYER 6 FEDERATION

Inter-jurisdictional protocol enabling sovereign nodes to share proofs while maintaining data sovereignty. No central authority.

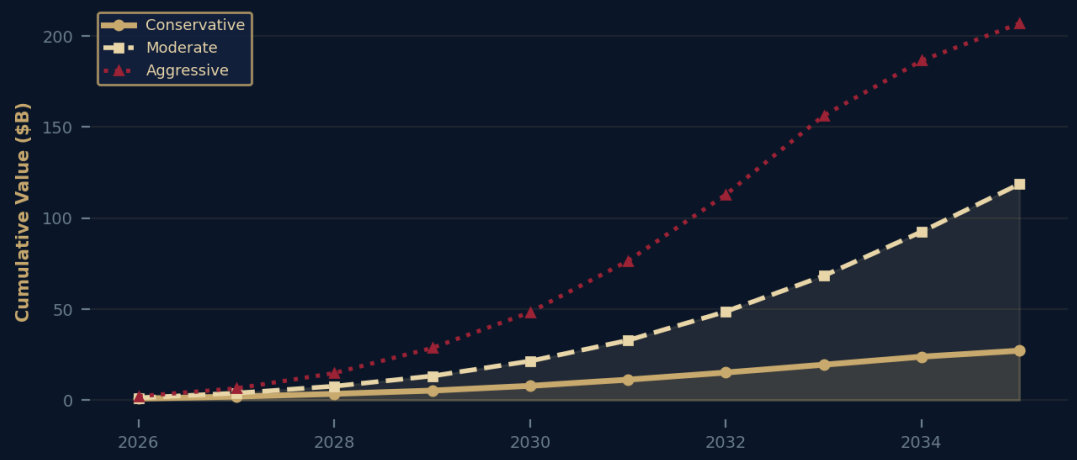
ARCHITECTURE CAPABILITY ASSESSMENT



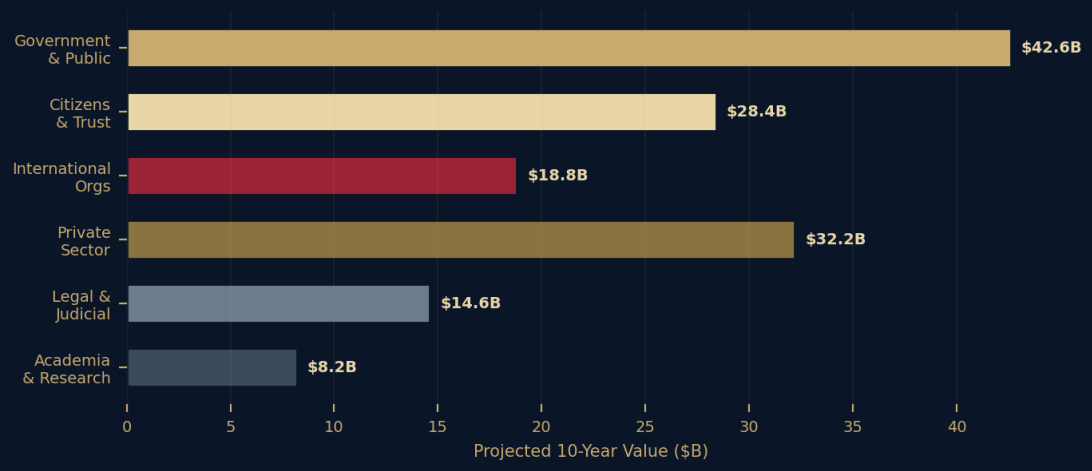
Value Proposition & ROI Analysis

GRGF creates systemic value by reducing ambiguity in institutional decision-making, increasing trust through verifiable records, and protecting honest actors with non-repudiable proof of diligent action. Value is measured across six stakeholder categories spanning 120+ direct and indirect applications.

10-YEAR CUMULATIVE VALUE TRAJECTORY ($B)



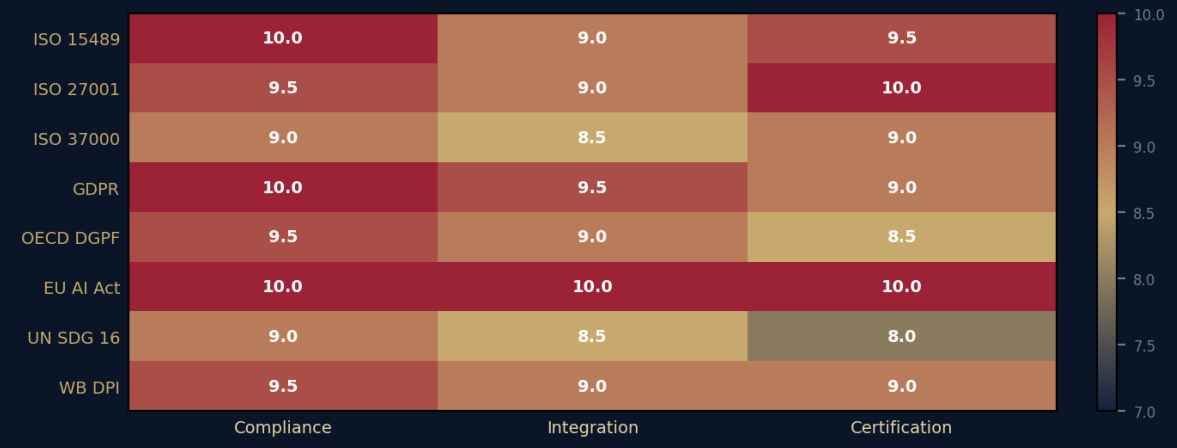
STAKEHOLDER VALUE DISTRIBUTION — 10-YEAR PROJECTION



International Standards Compliance

GRGF maps to 12+ international standards and regulatory frameworks. The deterministic design enables formal non-applicability certification for the EU AI Act (no probabilistic inference, no training data dependency). Full compliance matrices are provided for ISO 15489, ISO 27001, ISO 37000, GDPR, OECD Privacy Guidelines, and more.

STANDARDS COMPLIANCE MATRIX — ASSESSMENT SCORES



ISO 15489	Records Management — full alignment with lifecycle management, metadata, and disposition requirements.
ISO 27001	Information Security — certifiable compliance for ISMS including access control, cryptographic controls, and incident response.
ISO 37000	Governance of Organizations — guiding principles alignment for accountability, transparency, and stakeholder engagement.
GDPR (EU)	Data Protection — rights mapping including right to explanation, data portability, and purpose limitation by design.
OECD DGPF	Digital Government Policy Framework — complete compliance with openness, engagement, and data-driven principles.
EU AI Act	Non-Applicability Certification — deterministic system with zero probabilistic inference. Formal exemption documented.
UN SDG 16	Peace, Justice & Strong Institutions — direct contribution through transparent, accountable governance infrastructure.
World Bank DPI	Digital Public Infrastructure — alignment as sovereign trust-layer component with federation capabilities.

Sovereign Deployment Roadmap

PHASE 1: FOUNDATION (Q1–Q2 2026)

- Establish IP protection and legal framework (Canadian Patent CA 3,300,102)
- Complete institutional documentation suite (4 volumes, trilingual)
- Deploy digital presence and stakeholder engagement platform
- Initiate conversations with Government of Canada (PSPC, TBS)

PHASE 2: CANADA PILOT (Q3–Q4 2026)

- Launch pilot with 2-3 federal departments (Procurement, Health, Immigration)
- Deploy Six-Layer Architecture in controlled sovereign environment
- Validate deterministic governance logic with real institutional workflows
- Conduct independent third-party security and compliance audit

PHASE 3: SCALE & CERTIFY (2027)

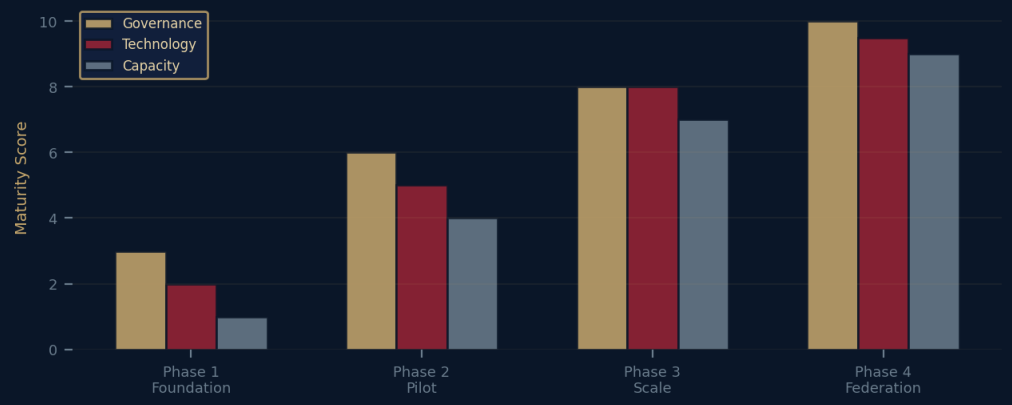
- Expand to 10+ departments and provincial governments
- Launch GRGF Academy with 4-tier certification program (76+ hours)
- Achieve ISO 27001 and ISO 15489 formal certification
- Begin international stakeholder engagement (World Bank, OECD, UN)

PHASE 4: GLOBAL FEDERATION (2028–2035)

- Deploy federation protocol enabling sovereign inter-jurisdictional proof sharing
- Establish GRGF as DPI trust-layer standard in G20 digital governance
- Scale to 50+ sovereign nodes across 6 continents
- Achieve $27.1B+ cumulative value in conservative scenario

Maturity Model & Governance Authority

GOVERNANCE MATURITY PROGRESSION BY PHASE



ANTI-CAPTURE CLAUSES

- AC-01: No single entity — government, corporation, or individual — may alter the core deterministic logic of the framework.
- AC-02: All governance records are append-only. No retroactive modification, deletion, or silent amendment is permitted.
- AC-03: The framework must remain vendor-neutral. No proprietary technology may be required for core operation.
- AC-04: Inventor economic interest (40%) is protected to ensure long-term stewardship independence.
- AC-05: Framework governance decisions require transparent multi-stakeholder process with published rationale.

GOVERNANCE AUTHORITY MODEL

Framework Architect: Tarek Wahid — sole inventor, patent holder, and steward of deterministic governance logic.

Governance Board: Independent multi-stakeholder body with published charter, transparent decision-making, and public minutes.

Technical Committee: Open-source community-driven standards body for protocol evolution and interoperability.

Certification Authority: Accredited training providers operating under GRGF Academy 4-tier certification framework.

Historical What-If Analysis

Five landmark governance failures are analyzed to demonstrate the counterfactual value GRGF would have provided. Total documented losses: $113.2B. These cases prove that governance record infrastructure is not optional — it is critical.

2008 GLOBAL FINANCIAL CRISIS — $67.4B

Missing governance records enabled unauthorized risk accumulation across institutions. GRGF's append-only audit trails and omission-awareness would have flagged regulatory gaps 18-24 months before cascade failure.

IRAQ RECONSTRUCTION (2003–2011) — $21.8B

Documented fraud, waste, and mismanagement in reconstruction contracts. Deterministic procurement records with chain-of-custody verification would have prevented contract manipulation and fund diversion.

COVID-19 EMERGENCY PROCUREMENT — $12.6B

Fast-tracked procurement without adequate governance trails led to overpayment and fraud. GRGF enables rapid governance with full auditability — speed and accountability are not mutually exclusive.

EU STRUCTURAL FUNDS MISUSE — $8.2B

Cross-border fund misallocation due to fragmented governance records. Federation protocol would enable sovereign record sharing while maintaining jurisdictional data sovereignty.

PHOENIX PAY SYSTEM (CANADA) — $3.2B

Failed IT transformation with no governance record of decision rationale. Record-of-omission framework would have documented missing approvals and flagged procedural gaps before deployment.

Academy, Training & Certification

The GRGF Academy establishes a 4-tier professional certification model totaling 76+ structured training hours. Each level builds upon the previous, creating a pipeline from foundational understanding to full architectural mastery.

LEVEL 1: FOUNDATIONS

16 Hours

Core concepts, governance record types, deterministic logic principles, and ethical framework.

LEVEL 2: PRACTITIONER

24 Hours

Implementation methodology, API integration, policy-as-code (OPA/Rego), and operational procedures.

LEVEL 3: SPECIALIST

20 Hours

Advanced cryptographic protocols, federation architecture, compliance mapping, and audit methodology.

LEVEL 4: ARCHITECT

16 Hours

System design, sovereign deployment planning, anti-capture governance, and strategic advisory.

SIMULATION LABS & ASSESSMENTS

- 10 hands-on simulation lab exercises covering real-world governance scenarios
- Professional Code of Ethics with 5 Articles governing certified practitioners
- Comprehensive assessment frameworks with written and practical examinations
- Governance Glossary with 200+ standardized terms and definitions
- Continuing education requirements for certification maintenance

APPLICATIONS MATRIX

- Public Procurement — Verifiable contract trails preventing manipulation and ensuring fair competition
- Healthcare Governance — Patient record integrity, clinical trial audit, and regulatory compliance
- Land Registry — Immutable title records with chain-of-custody for ownership transfers
- Electoral Oversight — Transparent ballot chain-of-custody and result verification
- Crisis Response — Rapid governance with full auditability for emergency procurement
- Judicial Records — Court proceeding integrity and evidence chain management
- Financial Regulation — Systemic risk monitoring through governance event transparency
- Environmental Compliance — Permit tracking, emissions reporting, and inspection records

Intellectual Property & Legal Framework

PATENT REGISTRATION

Canadian Patent No. CA 3,300,102

Inventor & Sole Owner: Tarek Wahid

Inventorship encompasses: Conceptual architecture, governance logic, record-of-omission framework, DPI trust-layer design, Evidence Backbone methodology, federation protocol design.

PROTECTED ELEMENTS

Architecture System architecture, six-layer model, Evidence Backbone, deterministic processing pipeline.

Logic Record types, event models, governance logic, omission detection algorithms.

Framework Authority framework, anti-capture mechanisms, stewardship governance model.

Verification Audit concepts, positive/negative verification, Merkle transparency log protocols.

Methodology Deterministic governance approach, policy-as-code paradigm, federation protocol.

LICENSING PRINCIPLES

- GRGF may be licensed for public interest use under terms preserving governance neutrality and inventor attribution.
- Commercialization does not compromise governance neutrality — economic sustainability ensures long-term independence.
- Inventor economic interest (40%) protected under Anti-Capture Clause AC-04 to prevent stewardship erosion.
- All derivative works, deployments, and institutional references must include mandatory attribution.
- Controlled disclosure protocols govern access to restricted materials — disclosure is governed, not discretionary.

Trust increases when records exist, omissions are visible, and authority is explicit.

GRGF restores trust by design, not messaging.

INVENTOR & FRAMEWORK ARCHITECT

Tarek Wahid

Canadian Patent No. CA 3,300,102 · Sovereign Authority Edition · 2026